

JPE Replication Report

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1 Paper Identification

ID	20241434
Author	Ganapati
Title	From Heterogeneous Firms to Heterogeneous Trade Elasticities: The Aggregate Implications of Firm Export Decisions
Iteration	2

Some useful information up front:

- [JPE Data and Code Availability Policy](#)
- [Step by step guidance from Data Editor for Authors](#)
- [Template README](#)

2 Summary

In assessing compliance with our [Data and Code Availability Policy](#), our reproducibility team has identified the following issues, which we ask you to address:

1. My replicator did a great job fixing a but in the data creation code (or so we think). Please check.
2. Our matlab application crashed several times. We are asking for your assistance in providing a smaller scale version of your monte carlo experiments, because we think we run out of memory. Of course those tests would not have to replicate the results in the paper.
3. There are a few remaining discrepancies. Please check.

3 General

- ☒ Data Availability and Provenance Statements
 - ☒ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
 - ☒ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
 - ☐ License for Data (optional, but recommended)
 - ☒ Details on each Data Source
- ☒ Dataset list
- ☒ Computational requirements
 - ☒ Software Requirements
 - ☒ Controlled Randomness (as necessary)
 - ☒ Memory, Runtime, and Storage Requirements
- ☒ Description of programs/code
 - ☐ License for Code (Optional, but recommended)
- ☒ {REPLICATOR} to Replicators
 - ☐ Details (as necessary)
- ☒ List of tables and programs
- ☒ References (Optional, but recommended)

Comment: My compliments on the changes to the README, it definitely improved a lot. Especially the dependency path is great and a good idea for a complicated package like this.

4 High-level Description of Research Project

This research project...

- ☒ uses observational data.
- ☒ uses *confidential* observational data.
- ☐ uses data generated via experiment or trial by the authors (in field or lab).
 - ☐ For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.

- ☐ A PDF copy of the instructions given to participants in both original language and an English translation is included.
- ☐ uses data generated via survey by the authors (in field or online)
 - ☐ The programs used to administer the survey (e.g. qualtrics survey file) are included.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF of the original questionnaire(s) and information on the selection and eligibility of participants is included.
- ☐ is a simulation study: the data are generated by an algorithm.
- ☐ is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

5.1 Data Sources

Data Source / file name	public	private	DOI/URL	Access described	cited in README	cited in paper
UNCTAD TRAINS – Tariff Data	no	no	yes	yes	yes	yes
World Bank EDD – Exporter Dynamics Database	yes	no	yes	yes	yes	yes
World Bank EDD – Colombian Microdata	no	yes	no	yes	yes	yes
CEPII – Gravity Variables	yes	no	yes	yes	yes	yes
WIOD – World Input-Output Database (2016)	yes	no	yes	yes	yes	yes
Teti Tariffs – Feodora Teti Data	yes	no	yes	yes	yes	yes
OECD SDBS/TEC – Firm Demographics & TEC	yes	no	yes	yes	yes	yes
US Census – Manufacturer Enterprises (2012)	yes	no	yes	yes	yes	yes
Australia Data – Characteristics of Exporters	yes	no	yes	yes	yes	yes
China Statistical Yearbook (2019)	yes	no	yes	yes	yes	yes
China Export Data	no	yes	no	yes	yes	yes
World Bank Enterprise Surveys	yes	no	yes	yes	yes	yes
GSP Rates – Constructed from TRAINS	no	no	no	yes	yes	yes
Eora MRIO – Multi-Region Input-Output Data	yes	no	yes	yes	yes	yes

6 Requirements

- ☒ README is in TXT, MD, or PDF format
- ☒ README is accessible at the root of the package (not inside a folder)
- ☒ Title conforms to guidance (starts with “Data and Code for: Title of Paper” or “Code for: Title of Paper”, is properly capitalized)
- ☒ Authors (with affiliations) are listed in the same order as on the paper

6.1 Stated Requirements

- ☐ No requirements specified
- ☒ Software Requirements specified as follows:
 - Stata 16.1/MP
 - Matlab 2023a
- ☒ Computational Requirements specified as follows:
 - OS X environment (M2 processor with 16 cores and 32 GB) and a 1 TB SSD for all the programs except `run\_mc\_master` which was run on an external Linux cluster with 100 available cores (32 GB addressable each). Programs do require z-shell scripting, so they do need an *nix/OSX environment.
- ☒ Time Requirements specified as follows:
 - 8-24 hours
- ☒ Requirements are complete.

7 Analyzing Data and Code Files

7.1 Potential Personal Identifiable Information (PII)

⚠ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals. If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

Summary: - Data files with PII indicators: 54 - Variables flagged in data: 171 - Code files with PII references: 76 - PII references in code: 1180

7.1.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	AUS_cleaned.csv	1	country
Data	COL_EXP_2012.dta	2	country
Data	CYD_manuf.dta	1	country
Data	CYH2D_manuf.dta	2	country
Data	CYH2D_manuf_1000usd.dta	2	country
Data	CY_manuf.dta	1	country
Data	Combined_Industrial_Enterprises_Indicators.xlsx	1	son
Data	Countries_V202102.dta	4	country, name, lon
Data	EDDVariableNames.xlsx	1	name
Data	European_Union.dta	1	country
Data	Full_Trade.csv	1	lat
Data	Gravity_V202102.dta	19	country, lat, lon
Data	SDBS_BDI_ISIC4_11042019204932050.csv	2	loc, location, country
Data	SSIS_BSC_ISIC4_11042019175351230.csv	2	loc, location, country
Data	Simulated_data_LogNormal_alt.mat	2	lon
Data	TES3_04052018220827756.csv	2	country
Data	WIOT2000_October16_ROW.dta	1	country
Data	WIOT2001_October16_ROW.dta	1	country
Data	WIOT2002_October16_ROW.dta	1	country
Data	WIOT2003_October16_ROW.dta	1	country
Data	WIOT2004_October16_ROW.dta	1	country
Data	WIOT2005_October16_ROW.dta	1	country
Data	WIOT2006_October16_ROW.dta	1	country
Data	WIOT2007_October16_ROW.dta	1	country
Data	WIOT2008_October16_ROW.dta	1	country
Data	WIOT2009_October16_ROW.dta	1	country
Data	WIOT2010_October16_ROW.dta	1	country
Data	WIOT2011_October16_ROW.dta	1	country
Data	WIOT2012_October16_ROW.dta	1	country
Data	WIOT2013_October16_ROW.dta	1	country
Data	WIOT2014_October16_ROW.dta	1	country
Data	aff_download_readme_ann.txt	2	zip, lat
Data	country_codes_V202201.csv	3	country, name

File Type	File	Variables/References	PII Categories
Data	dist_cepii.dta	7	lat, lon, country
Data	dist_cepii.xls	1	lon
Data	dist_cepii_bra.dta	7	lat, lon, country
Data	eu_exporters.dta	1	country
Data	eu_importers.dta	1	country
Data	exports_2007.dta	1	country
Data	exports_2008.dta	1	country
Data	exports_2009.dta	1	country
Data	exports_2010.dta	1	country
Data	exports_2011.dta	1	country
Data	exports_2012.dta	1	country
Data	fpart_concordance.dta	1	country
Data	geo_cepii.dta	25	country, name, loc, city, lat, degree, lon, son
Data	geo_cepii.xls	15	country, loc, city, lat, lon
Data	geo_cepii_maincity.dta	25	country, name, loc, city, lat, degree, lon, son
Data	gravdata_cepii.dta	13	lat, lon
Data	index_readme.txt	1	block, loc
Data	index_readme.txt	1	block, loc
Data	tariff2010_beta1-2024-12.dta	1	name
Data	tariff2012_beta1-2024-12.dta	1	name
Data	tariff2014_beta1-2024-12.dta	1	name
Code	AppendixFigure_LogPareto.m	4	lon, loc, location, lat
Code	CF_agg_plot.m	1	lat
Code	CF_diagnostic.m	3	lat, loc, location
Code	CF_flow_plot.m	2	loc, location, lat
Code	Figure_Sequencer.m	5	city, lon
Code	GFT.m	7	lat, lon
Code	GFT.m	7	lat, lon
Code	GFTG_v10.m	4	lon
Code	GFTG_v10.m	4	lon
Code	GFT_Script_v17_EKK.m	13	lon, lat, city, loc, location
Code	GFT_agg_plot.m	2	lat, loc, location
Code	GFT_division.m	1	city, loc, location
Code	GFT_division2.m	1	city, loc, location
Code	GFT_fct_general_fast.m	27	lon, lat, city
Code	GFT_pre.m	4	lat, lon

File Type	File	Variables/References	PII Categories
Code	GFT_pre1.m	4	lat, lon
Code	GFT_script_v17.m	134	lon, city, lat, name, loc, location
Code	GFT_v9.m	5	lon
Code	GFT_v9.m	5	lon
Code	GMM_gravity.m	3	lon, second
Code	GMM_gravity.m	5	lon, second
Code	GMM_gravity_log_normal.m	6	lon, lat, second
Code	GMM_wrapper_gravity.m	20	lon, dob, second
Code	GMM_wrapper_gravity.m	21	lon, name, dob, second
Code	GenerateBaseline.m	3	name, lon
Code	Graphs_GSP.m	21	city, name, lat, loc, location
Code	Graphs_TC.m	33	city, name, lat, loc, location
Code	LinearCF.m	20	second, name, lon, lat
Code	LinearCF_wrapper_v3.m	1	lat
Code	MasterProgram.do	7	name, loc, lat
Code	Master_script_gravity_v3.m	79	lon, name, city, lat, loc, location
Code	QuantilesBaseline.m	1	lat
Code	QuantilesColombia.m	1	dob
Code	Quantiles_Colombia_Wrapper.m	10	name, country
Code	Script_GMM_simulation.m	28	lat, lon, loc, name, second
Code	Script_QQ_simulation.m	27	lat, lon, loc, name, second, city
Code	Simulation_Data_v10.m	51	lat, city, lon, loc, location, name
Code	Test_Code.m	2	lon
Code	create_elasticity_functions.m	19	city, lon, lat, son
Code	figure_plotter_QQ_minimal.m	9	lat, degree, name
Code	figures_gen.m	41	city, lat, son, name, loc, location, child, dob
Code	function_interp_wrapper.m	9	lon, city, lat, son
Code	function_interp_wrapper_GMM.m	8	lon, city
Code	generate_QQ_plot.m	19	lat
Code	generate_eps.m	1	lon
Code	generate_pdf_entry.m	5	lat, name, loc, location
Code	get_fct_form_params.m	4	city
Code	graph_overlay.m	31	lon, lat, loc, location, city, name
Code	hs_routine_gravity.m	8	lon, city

File Type	File	Variables/References	PII Categories
Code	inv_unique_interp.m	1	lat
Code	logpareto_inv_shifted.m	3	loc, location, lat
Code	make_data_combo.m	19	lon, name, lat, son
Code	make_data_combo_hs_multi.m	14	lon, name, lat
Code	make_data_simulation.m	3	lat, name, son
Code	make_elasticity_combo.m	3	city, lon
Code	p01_TRAINS_clean_data_v1_AAG.do	52	country, name, zip, loc, lat, lon
Code	p02_ChinaData_Morrow.do	3	loc, country, name
Code	p02_eora_trade_matrix_clean.do	9	name, lon
Code	p02_quantiles_colombia_firmexports.do	1	name
Code	p03_Create_Tariffs.do	6	name
Code	p04_Sample_Creation.do	129	name, country, lon, loc, lat, location, census, sex
Code	p05_Sample_Creation_HS2.do	9	name, lon
Code	p06_ReducedForm.do	7	lon, loc
Code	p_overlay_elasticity.m	101	city, name, lon, lat, loc, location
Code	plot_Er_vs_log_n.m	3	name, lon, loc, location
Code	plot_logcorr_pareto.m	1	lon
Code	plot_theta_c_vs_log_n.m	2	name, loc, location
Code	plot_theta_combined.m	6	lon, name, loc, location
Code	plot_theta_e_vs_log_n.m	9	lon, lat, name, loc, location
Code	plot_theta_i_vs_log_n.m	6	lon, name, loc, location
Code	recover_elasticities_LN.m	6	lon, lat
Code	regQQ.m	4	lon, lat
Code	run_mc_master.m	34	lat, city, lon, name
Code	solve_FP.m	19	lat, city, lon, country, name
Code	vline.m	2	loc, location, lat
Code	vline.m	2	loc, location, lat

See [Appendix](#) for detailed listing of all flagged instances.

7.1.2 File Paths Report

Generated on 2026-03-23T11:39:32.429

Warning: Our search on file path types is imperfect and incurs both type 1 and type 2 errors. We aim to strike a reasonable balance between both. The below table is therefore only indicative. Detailed listings can be found in the appendix to this report.

In this table we analyse all files which contain source code (not data and documentation), and we report the grand total of each type of filepath we encountered.

Please check and replace any `windows` filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that `STATA` allows `/` to be a filepath separator on a windows platform, hence this is a requirement for *all* `STATA` applications.)

Files Analyzed	Windows Paths	Unix Paths	Mixed Paths	Drive Letters (C:\ etc)
98	2	30	7	0

7.1.3 Duplicate Files Report

We found the following duplicate files:

Filename	Size (MB)	Checksum (MD5)
Data/EORA/Raw_data/Eora26_2012_bp/labels_VA.txt	0.0	3687db9cd4cecf9fd396ee0045
Code/MonteCarlo/functions/evknots_make_deriv.m	0.0	47df39afafbdb1503a71b5fd59bc
Data/EORA/Raw_data/Eora26_2012_bp/index_readme.txt	0.0	938c04a07c9575c041e92527bfc
Code/MonteCarlo/functions/evknots_make.m	0.0	31974cba393db94eecbcd9a5cee
Code/MonteCarlo/functions/GFT_v2/GFTG_v10.m	0.0	bbfe609423a0fbc935db72314e
Code/MonteCarlo/functions/GFT_v2/GFT_v9.m	0.0	3e91184a622cf9c9026d6989aa
Code/MonteCarlo/functions/GFT_v2/GFT.m	0.0	6f8380bcb7fd0162b0bfaa16144e
Code/MonteCarlo/functions/vline.m	0.0	91f059f68599452e21a3a7fdb83
Data/EDD_Data/Dictionary/TechnicalNote/Appendix.xlsx	0.05	09acce97fb1cafe1e265f8c4fd0f5
Data/EORA/Raw_data/Eora26_2012_bp/labels_FD.txt	0.07	5d74450d3ac9b8cc77d9d3ad4b
Data/EORA/Raw_data/Eora26_2012_bp/labels_Q.txt	0.15	0c273fed165f04eacca0dc8ffca6
Data/EORA/Raw_data/Eora26_2012_bp/labels_T.txt	0.21	ce21a96f119a9df418e6bb014229
Data/EORA/Raw_data/Eora26_2012_bp/Eora26_2012_bp_VA.txt	0.37	d523cd2a7499f104e3e6078cdd3
Data/EORA/Raw_data/Eora26_2012_bp/import/miranda/Eoras/Phase199/Loop082/simplified/Eora26_2012_bp_VA.txt	0.37	d523cd2a7499f104e3e6078cdd3
Data/EORA/Raw_data/import/miranda/Eoras/Phase199/Loop082/simplified/Eora26_2012_bp_VA.txt	0.37	d523cd2a7499f104e3e6078cdd3
Data/EDD_Data/Dictionary/TechnicalNote/ConsolidationNote_2015_.pdf	0.52	1e6a65233ed1877d8568c88bd5
Data/China_Statistical_Yearbook/Raw/China Statistical Yearbook 2019_files/E1302_002.jpg	0.61	6a04329e0426be7693de12c261
Data/EORA/Raw_data/Eora26_2012_bp/Eora26_2012_bp_QY.txt	38.21	58294a35fa8d88b0754ef750e7d
Data/EORA/Raw_data/Eora26_2012_bp/Eora26_2012_bp_FD.txt	68.85	7d15c8bc188563ed26adc8ef677
Data/EORA/Raw_data/Eora26_2012_bp/import/miranda/Eoras/Phase199/Loop082/simplified/Eora26_2012_bp_FD.txt	68.85	7d15c8bc188563ed26adc8ef677
Data/EORA/Raw_data/import/miranda/Eoras/Phase199/Loop082/simplified/Eora26_2012_bp_FD.txt	68.85	7d15c8bc188563ed26adc8ef677
Data/EORA/Raw_data/Eora26_2012_bp/Eora26_2012_bp_Q.txt	164.76	f6d883acce2ebc89c8dfed59c8a
Data/EORA/Raw_data/Eora26_2012_bp/Eora26_2012_bp_T.txt	296.84	c1b90cf89b5fc6b223d10c38363
Data/EORA/Raw_data/Eora26_2012_bp/import/miranda/Eoras/Phase199/Loop082/simplified/Eora26_2012_bp_T.txt	296.84	c1b90cf89b5fc6b223d10c38363
Data/EORA/Raw_data/import/miranda/Eoras/Phase199/Loop082/simplified/Eora26_2012_bp_T.txt	296.84	c1b90cf89b5fc6b223d10c38363

7.1.4 Zero Size Files Report

We did not find any zero sized files.

7.1.5 Large Files Report

We found the following files larger than 100MB:

Filename	Size (MB)
Data/EORA/Raw_data/Eora26_2012_bp_Q.txt	164.76
Data/EORA/Raw_data/Eora26_2012_bp/Eora26_2012_bp_Q.txt	164.76

Filename	Size (MB)
Data/EORA/Raw_data/Eora26_2012_bp_T.txt	296.84
Data/EORA/Raw_data/Eora26_2012_bp/Eora26_2012_bp_T.txt	296.84
Data/EORA/Raw_data/Eora26_2012_bp/import/miranda/Eoras/Phase199/Loop082/simplified/Eora26_2012_bp_T.txt	296.84
Data/EORA/Raw_data/import/miranda/Eoras/Phase199/Loop082/simplified/Eora26_2012_bp_T.txt	296.84
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2000_V202201.csv	354.76
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2001_V202201.csv	366.9
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2002_V202201.csv	376.96
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2003_V202201.csv	389.46
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2004_V202201.csv	407.54
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2005_V202201.csv	422.08
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2006_V202201.csv	438.47
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2007_V202201.csv	449.43
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2009_V202201.csv	449.46
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2008_V202201.csv	455.21
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2010_V202201.csv	462.52
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2011_V202201.csv	470.89
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2012_V202201.csv	484.09
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2013_V202201.csv	493.35
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2014_V202201.csv	495.26
Data/CEPII/Gravity/gravdata_cepil.dta	581.64
Data/EDD_Data/CYH2D_manuf/CYH2D_manuf_1000usd.dta	810.31
Data/EDD_Data/CYH2D_manuf/CYH2D_manuf.dta	814.36
Data/CEPII/Gravity_dta_V202102/Gravity_V202102.dta	1183.59
Data/TRAINS/raw_data/DataJobID-1788312_1788312_TRAINS1995.dta	2249.73
Data/TRAINS/raw_data/DataJobID-1788495_1788495_TRAINS1996new.dta	3238.57
Data/TRAINS/raw_data/DataJobID-1788325_1788325_TRAINS1998.dta	3295.35
Data/TRAINS/raw_data/DataJobID-1788317_1788317_TRAINS1997.dta	3599.68
Data/TRAINS/raw_data/DataJobID-1788353_1788353_TRAINS1999.dta	3997.68
Data/TRAINS/raw_data/DataJobID-1788359_1788359_TRAINS2000.dta	5245.66
Data/Teti/tariff2010_beta1-2024-12.dta	5658.99
Data/Teti/tariff2012_beta1-2024-12.dta	5658.99
Data/Teti/tariff2014_beta1-2024-12.dta	5658.99
Data/TRAINS/raw_data/DataJobID-1788462_1788462_TRAINS2001.dta	6041.71
Data/TRAINS/raw_data/DataJobID-1788466_1788466_TRAINS2003.dta	6328.54
Data/TRAINS/raw_data/DataJobID-1788463_1788463_TRAINS2002.dta	6467.25
Data/TRAINS/raw_data/DataJobID-1788496_1788496_TRAINS2004.dta	6545.66
Data/TRAINS/raw_data/DataJobID-1788497_1788497_TRAINS2005.dta	7280.98
Data/TRAINS/raw_data/DataJobID-1788498_1788498_TRAINS2006.dta	7574.18

Filename	Size (MB)
Data/TRAINS/raw_data/DataJobID-1788548_1788548_TRAINS2014.dta	7641.07
Data/TRAINS/raw_data/DataJobID-1791395_1791395_TRAINS2017.dta	7840.27
Data/TRAINS/raw_data/DataJobID-1788508_1788508_TRAINS2007.dta	7873.51
Data/TRAINS/raw_data/DataJobID-1788516_1788516_TRAINS2010.dta	7909.5
Data/TRAINS/raw_data/DataJobID-1788515_1788515_TRAINS2009.dta	7936.12
Data/TRAINS/raw_data/DataJobID-1788563_1788563_TRAINS2018.dta	8002.6
Data/TRAINS/raw_data/DataJobID-1788547_1788547_TRAINS2012.dta	8131.58
Data/TRAINS/raw_data/DataJobID-1788514_1788514_TRAINS2008.dta	8234.33
Data/TRAINS/raw_data/DataJobID-1788524_1788524_TRAINS2011.dta	8375.05
Data/TRAINS/raw_data/DataJobID-1788554_1788554_TRAINS2013.dta	8505.74
Data/TRAINS/raw_data/DataJobID-1788549_1788549_TRAINS2015.dta	8635.54
Data/TRAINS/raw_data/DataJobID-1788561_1788561_TRAINS2016.dta	8714.73

7.2 Code description

- There are eight Stata do-files, six Matlab .m files and one master do-file (Stata and Matlab combined).

cloc github.com/AIDanial/cloc v 2.06 T=6.94 s (18.4 files/s, 99683.1 lines/s)				
Language	files	blank	comment	code
CSV	13	0	0	659509
Text	13	104	0	16765
MATLAB	83	1769	1433	6597
HTML	8	132	61	2612
Stata	9	574	204	2051
Markdown	1	60	0	225
XML	1	0	0	1
---	---	---	---	---
SUM:	128	2639	1698	687760

☒ The replication package contains a “main” or “master” file(s) which calls all other auxiliary programs.

NOTE: In-text numbers that reference numbers in tables do not need to be listed. Only in-text numbers that correspond to no table or figure need to be listed.

The table below provides an overview of all intermediate and final datasets generated throughout the replication. This information is provided in the README.

Dataset	Description	Generated By	Used By
[YEAR]_Teti <u>uw</u> .csv	Baseline gravity sample (bilateral trade, tariffs from Teti, gravity controls)	p04_Sample_Creation.do	Master_script_gravity_v3.m, GFT_script_v17.m, LinearCF_wrapper_v3.m
[YEAR]_Tu <u>w</u> .csv	Alternative gravity sample (tariffs from TRAINS)	p04_Sample_Creation.do	Master_script_gravity_v3.m
[YEAR]_Tu <u>w</u> IV2.csv	IV specification sample	p04_Sample_Creation.do	Master_script_gravity_v3.m

Dataset	Description	Generated By	Used By
n_ij_[YEAR]B.csv	Bilateral number of exporters matrix (CxC)	p04_Sample_Creation.do	GFT_script_v17.m , LinearCF_wrapper_v3.m
x_ij_[YEAR]B.csv	Bilateral average export size matrix (CxC)	p04_Sample_Creation.do	GFT_script_v17.m
XX_ij_[YEAR]B.csv	Bilateral total exports matrix (CxC)	p04_Sample_Creation.do	GFT_script_v17.m , LinearCF_wrapper_v3.m
balance_i_[YEAR]B.csv	Country-level aggregates (expenditure, production, trade balance)	p04_Sample_Creation.do	GFT_script_v17.m
G_ij_[YEAR]B.csv	Bilateral group classification (developed/developing origin)	p04_Sample_Creation.do	GFT_script_v17.m , LinearCF_wrapper_v3.m
G4_ij_[YEAR]B.csv	Four-way group classification	p04_Sample_Creation.do	GFT_script_v17.m
Sample_[YEAR]B.csv	Sample selection indicators	p04_Sample_Creation.do	GFT_script_v17.m , LinearCF_wrapper_v3.m
l_i_[YEAR]B.csv	Country labels	p04_Sample_Creation.do	GFT_script_v17.m
N_ii_[YEAR]B.csv	Domestic firm counts	p04_Sample_Creation.do	LinearCF_wrapper_v3.m
GSP_delta_[YEAR]B.csv	GSP tariff reduction indicators	p04_Sample_Creation.do	LinearCF_wrapper_v3.m
GSP_[YEAR]B.csv	GSP eligibility indicators	p04_Sample_Creation.do	LinearCF_wrapper_v3.m
mfn_rate_[YEAR]B.csv	MFN tariff rates	p04_Sample_Creation.do	LinearCF_wrapper_v3.m
[YEAR]_survival_.csv	Firm survival rates (1-year)	p04_Sample_Creation.do	Master_script_gravity_v3.m
[YEAR]_survival_all_.csv	Firm survival rates (all)	p04_Sample_Creation.do	Master_script_gravity_v3.m
[YEAR]_survival_alt3_.csv	Firm survival rates (3-year)	p04_Sample_Creation.do	Master_script_gravity_v3.m
[YEAR]_nonii_.csv	Sample dropping imputed domestic firms	p04_Sample_Creation.do	Master_script_gravity_v3.m
Data/Int/stack_data_B.dta	Stacked panel dataset (all years, bilateral pairs with tariffs and gravity)	p04_Sample_Creation.do	p06_ReducedForm.do
Data/Int/CEPII_Grav_temp.dta	Cleaned CEPII gravity variables	p04_Sample_Creation.do	p05_Sample_Creation_HS2.do
Data/Int/CEPII_rich.dta	Developed/developing country classification	p04_Sample_Creation.do	p05_Sample_Creation_HS2.do
Output/[YEAR]_Sample_summary.tex	Estimation sample summary statistics	p04_Sample_Creation.do	Paper Table OA.2
Output/Hist_[YEAR]B.pdf	Histogram of exporter firm shares	p04_Sample_Creation.do	Paper Figure OA.6

Dataset	Description	Generated By	Used By
<code>gamma_gravity.mat</code>	Estimated gravity coefficients	<code>Master_script_gravity_v3.m</code>	<code>GFT_script_v17.m</code> , <code>LinearCF_wrapper_v3.m</code>
<code>ESTIMATES_LINEAR.mat</code>	Linear counterfactual estimation objects	<code>LinearCF_wrapper_v3.m</code>	<code>Graphs_TC.m</code> , <code>Graphs_GSP.m</code>
<code>ESTIMATES_SPLINE2.mat</code>	Spline (2 knots) counterfactual estimation objects	<code>LinearCF_wrapper_v3.m</code>	<code>Graphs_TC.m</code> , <code>Graphs_GSP.m</code>
<code>ESTIMATES_SPLINE4.mat</code>	Spline (4 knots) counterfactual estimation objects	<code>LinearCF_wrapper_v3.m</code>	<code>Graphs_TC.m</code> , <code>Graphs_GSP.m</code>
<code>ESTIMATES_SPLINE6.mat</code>	Spline (6 knots) counterfactual estimation objects	<code>LinearCF_wrapper_v3.m</code>	<code>Graphs_TC.m</code> , <code>Graphs_GSP.m</code>
<code>Output/Decomposition_Results_TC[scenario].csv</code>	Welfare decomposition results (tariff cut scenarios)	<code>Graphs_TC.m</code>	Paper Table 1
<code>Output/Decomposition_Results_GSP[scenario].csv</code>	Welfare decomposition results (GSP scenarios)	<code>Graphs_GSP.m</code>	Paper Table OA.6
<code>GFT_results_revision_2012B.xlsx</code>	Welfare decomposition results (GFT analysis)	<code>GFT_script_v17.m</code>	Paper Figure 5
<code>Simulated_data_[spec]_alt.mat</code>	Simulated economies (LogNormal, MPareto, MEstimates)	<code>solve_FP.m</code>	<code>Script_GMM_simulation.m</code> , <code>Script_QQ_simulation.m</code>
<code>Simulated_[spec]_Estimates_[basis]_[knots]_alt.mat</code>	GMM simulation estimates	<code>Script_GMM_simulation.m</code>	<code>figures_gen.m</code>
<code>Simulated_[spec]_Estimates_QQLogNormal_1_[force_n]_alt.mat</code>	QQ simulation estimates	<code>Script_QQ_simulation.m</code>	<code>figures_gen.m</code>

7.3 Computing Environment of the Replicator

- Nuvolos 16vCPU 64GB
- Stata/MP 16.0
- Matlab R2023a

8 Replication

8.1 Stata

- Downloaded code and data from dropbox.
- unzipped all the necessary datafiles.
- Ran the master do-file as per README, in Stata. Master do-file did not run smoothly in one go, I had to make some minor changes:

8.1.1 p04_Sample_Creation.do

- The authors said they found the issue in the block between lines 584 and 632 and solved it. However, I am now experiencing a new error `__000001 not found`. For some years, no survival rate data exists in the OECD dataset. When the code filtered for a specific year using `keep if TIME == ${year}`, this resulted in a dataset with 0 observations. This caused Stata to crash with the error `__000001 not found`, because `egen mean()` can't run on an empty dataset. Additionally, `saveold` saved an empty file, which then caused the merge in `stack_data` to fail.

To fix this, I added a `count` check after the merges. If the dataset is empty, I create a placeholder dataset with the correct variable structure but 0 observations. If data exists, the `egen` averages are computed as normal. This solved the issue, the code ran fine afterwards.

```

count
if r(N) == 0 {
  clear
  set obs 1
  gen iso3_o = ""
  gen SurvivalRate1 = .
  gen SurvivalRate2 = .
  gen SurvivalRate3 = .
  gen SurvivalRate1_average = .
  gen SurvivalRate2_average = .
  gen SurvivalRate3_average = .
  drop in 1
}
else {
  egen SurvivalRate1_average = mean(SurvivalRate1)
  label var SurvivalRate1_average "Average OECD 1-year survival rate"
  egen SurvivalRate2_average = mean(SurvivalRate2)
  label var SurvivalRate2_average "Average OECD 2-year survival rate"
  egen SurvivalRate3_average = mean(SurvivalRate3)
  label var SurvivalRate3_average "Average OECD 5-year survival rate"
}

```

- Furthermore, the README states that all TRAIN zip files must be manually unzipped before starting the replication. However, the code apparently already performs the unzipping, so this instruction can either be removed or adjusted accordingly?

```

unzipfile "TRAINS`year`.zip", replace
local files : dir "`c(pwd)'" files "*TRAINS`year`*.dta"

```

8.2 Matlab

- Downloaded code and data from dropbox.
- unzipped all the necessary datafiles.
- Ran the scripts per README. Most of the scripts ran fine and without any error.

Master_script_gravity_v3.m: Code seem to run fine and was able to generate some output but crashed at some point. I tried to run in multiple times but Matlab constantly shuts down at some point. I ran this part with 16CPUs and 64GB memory. As a result, I can't located figures OA.13,14,15 & 16 in my output.

LinearCF_wrapper_v3.m: Code seems to run fine, however looking at the output it seems to generate only a few files. Figure 6, Table 1, and Figures OA.29–OA.31 and Table OA.6 could not be located.

run_mc_master.m ran with 16 CPUs and 64GB of memory but Matlab constantly breaks down at the run_gmm step due to insufficient memory. Therefore, I was not able to replicate this part of the replication.

9 Findings

9.1 Missing Requirements

- ☐ Software Requirements

[REQUIRED] Please amend README to contain complete requirements.

You can copy the section above, amended if necessary.

9.2 Data Preparation Code

- Code/p01_TRAINS_clean_data_v1_AAG.do ran without any error.
- Code/p02_quantiles_colombia_firmexports.do ran without any error.
- Code/p02_eora_trade_matrix_clean.do ran without any error.
- Code/p02_ChinaData_Morrow.do ran without any error.
- Code/p03_Create_Tariffs.do ran without any error.
- Code/p04_Sample_Creation.do broke down with described error, after a minor change it ran without any error.
- Code/p05_Sample_Creation_HS2.do ran without any error.
- Code/p06_ReducedForm.do ran without any error.
- Code/GMM_estimation_v3/Master_script_gravity_v3.m ran but breaks down at some point, does not generate all output.
- Code/GFT_v2/GFT_script_v17.m ran without any error.

- Code/LinearCF_v3/LinearCF_wrapper_v3.m ran without any error but did not generate all expected output; Figure 6, Table 1, and Figures OA.29–OA.33 and Table OA.6 could not be located.
- Code/ColombiaTests/Quantiles_Colombia_Wrapper.m ran without any error. Code/MonteCarlo/run_mc_master.m ran with 16 CPUs and 64GB of memory but Matlab constantly breaks down at the run_gmm step due to insufficient memory.
- Code/LogCorrected/plots.m ran without any error.
- Code/Appendix_LogCorrected/AppendixFigure_LogPareto.m ran without any error.

9.3 Tables and Figures

Figure/Table #	Program	Reproduced?
Figure 1	Matlab	Yes
Figure 2	Matlab	No (MC)
Figure 3	Matlab	Yes
Figure 4	Matlab	Yes
Figure 5	Matlab	Yes
Figure 6	Matlab	No
Table 1	Matlab	No
Figure OA.1	Matlab	Yes
Figure OA.2	Matlab	Yes
Figure OA.3	Matlab	No (MC)
Figure OA.4	Matlab	No (MC)
Figure OA.5	Matlab	No (MC)
Figure OA.6	Stata	Yes
Table OA.1	-	N/A
Table OA.2	Stata	No
Table OA.3	-	N/A
Figure OA.7	Matlab	Yes
Figure OA.8	Matlab	Yes
Figure OA.9	Matlab	Yes
Figure OA.10	Matlab	Yes
Figure OA.11	Matlab	Yes
Figure OA.12	Matlab	Yes
Figure OA.13	Matlab	No
Figure OA.14	Matlab	No
Figure OA.15	Matlab	No
Figure OA.16	Matlab	No
Figure OA.17	Matlab	Yes
Figure OA.18	Matlab	Yes
Figure OA.19	Matlab	Yes
Figure OA.20	Matlab	Yes
Figure OA.21	Matlab	Yes
Figure OA.22	Matlab	Yes

Figure/Table #	Program	Reproduced?
Figure OA.23	Matlab	Yes
Figure OA.24	Matlab	Yes
Figure OA.25	Matlab	Yes
Figure OA.26	Matlab	Yes
Figure OA.27	Matlab	Yes
Figure OA.28	Matlab	Yes
Figure OA.29	Matlab	No
Figure OA.30	Matlab	No
Figure OA.31	Matlab	No
Figure OA.32	Matlab	Yes
Figure OA.33	Matlab	Yes
Table OA.4	Stata	No
Table OA.5	Matlab	Yes
Table OA.6	Matlab	No

- Figures OA.13, OA.14, OA.15, OA.16, OA.29, OA.30, OA.31, Figure 6, Table 1, and Table OA.6 were not generated by the corresponding codes ([LinearCF_wrapper_v3.m](#) and [Master_script_gravity_v3.m](#)). Therefore, these outputs can't be verified in this replication round.
- The tables below were generated by the corresponding code but showed discrepancies in their output compared to the tables in the paper. See the explanations and screenshots for further details.
- Table 1: Output seems completely different than table in the paper.

JPE-Ganapati-20241434 > replication-package > Output >  Decomposition_Results_TC2.csv

```

1  Spline 4-Group      3.16925362 3.22221584 -0.06469685 0.02660762 0.11497955 -0.13002018 0.00016764
2  Developed Origins  3.97731808 3.77716166 -0.09033538 0.02583251 -0.19707055 0.46150112 0.00022872
3  Developing Origins 1.79628548 2.27931715 -0.02113488 0.02792459 0.64517839 -1.13506363 0.0006386
4

```

- Table OA.2: The tables are identical except for the number of destinations, where BFA (53 vs 52), CMR (87 vs 86), and KEN (112 vs 108),

```

21 BFA00528-7.998-1.5381.4882.26 \tabularnewline
22 BGD001278-7.008-2.0282.0881.42 \tabularnewline
23 BGR001458-6.748-2.2681.9881.98 \tabularnewline
24 BOL00618-7.158-1.7381.4581.72 \tabularnewline
25 BRA001568-7.928-0.8281.9281.44 \tabularnewline
26 BWA008578-9.258-3.2381.6682.91 \tabularnewline
27 CAN016478-4.788-0.1881.2981.31 \tabularnewline
28 CHL001358-5.138-1.3081.9081.59 \tabularnewline
29 CHN001558-4.478-1.1581.7380.99 \tabularnewline
30 CIV000978-8.418-0.9581.5281.72 \tabularnewline
31 CMR00868-7.798-2.0081.3781.51 \tabularnewline
32 COL001268-8.648-1.5682.0181.45 \tabularnewline
33 CRI001088-8.368-2.2181.9582.09 \tabularnewline
34 CYP010348-5.288-0.9781.1381.14 \tabularnewline
35 CZE00588-4.188-0.2181.0580.86 \tabularnewline
36 DEU010358-3.1580.4780.7481.15 \tabularnewline
37 DNK0101558-4.698-1.5581.6981.37 \tabularnewline
38 DOM001108-8.708-2.0981.6681.51 \tabularnewline
39 ECU001068-8.698-1.5381.7681.82 \tabularnewline
40 EGY001368-7.618-1.7281.5881.49 \tabularnewline
41 ESP0101578-5.788-1.7681.9581.28 \tabularnewline
42 EST000708-4.428-1.0081.2981.29 \tabularnewline
43 ETH000748-8.148-2.6881.2682.34 \tabularnewline
44 FIN010358-3.4880.3080.9381.20 \tabularnewline
45 FRA010588-3.9980.1780.9080.99 \tabularnewline
46 GAB00568-6.478-1.9881.3382.18 \tabularnewline
47 GBR010588-3.968-0.2080.8580.99 \tabularnewline
48 GEO00848-9.178-1.4381.6081.77 \tabularnewline
49 GIN00608-8.738-2.3181.3183.02 \tabularnewline
50 GRG010588-5.658-0.6181.2581.07 \tabularnewline
51 GTM001068-8.218-1.9281.9481.98 \tabularnewline
52 HRV001018-7.298-2.4981.7082.33 \tabularnewline
53 HUN000358-4.5880.6681.1580.91 \tabularnewline
54 IRL010358-3.6681.1580.9581.40 \tabularnewline
55 ITA010588-3.548-0.8981.1480.82 \tabularnewline
56 JOR0001188-7.048-2.0781.7281.71 \tabularnewline
57 KEN0001088-7.348-2.8081.7181.69 \tabularnewline
58 KGZ00578-7.588-2.0681.4282.03 \tabularnewline
59 KHM00938-7.258-2.5681.6281.92 \tabularnewline

```

Table 2

- Table OA.4: A lot of discrepancies in the generated output.

```

\begin{tabular}{l*{4}{c}}
\hline\hline
& \multicolumn{1}{c}{(1)} & \multicolumn{1}{c}{(2)} & \multicolumn{1}{c}{(3)} & \multicolumn{1}{c}{(4)} \\
& \multicolumn{1}{c}{Baseline} & \multicolumn{1}{c}{+ Controls} & \multicolumn{1}{c}{Split} & \multicolumn{1}{c}{+ Controls} \\
\hline
ln(Distw\_ij) & 1.315 & 1.289 & & \\
& (0.0549) & (0.0554) & & \\
[1em]
cut\_n\_ij=1 $\times$ ln(Distw\_ij) & 0.207 & 0.123 & & \\
& (0.0584) & (0.0599) & & \\
[1em]
rich\_orig=0 $\times$ ln(Distw\_ij) & & & 1.203 & 1.182 \\
& & (0.0640) & (0.0645) & \\
[1em]
cut\_n\_ij=1 $\times$ rich\_orig=0 $\times$ ln(Distw\_ij) & & & 0.454 & 0.377 \\
& & (0.0683) & (0.0696) & \\
[1em]
rich\_orig=1 $\times$ ln(Distw\_ij) & & & 1.416 & 1.353 \\
& & (0.0902) & (0.0911) & \\
[1em]
cut\_n\_ij=1 $\times$ rich\_orig=1 $\times$ ln(Distw\_ij) & & & -0.253 & -0.314 \\
& & (0.0965) & (0.0975) & \\
\hline
N & 9652 & 9652 & 9652 & 9652 \\
r2 & 0.799 & 0.801 & 0.801 & 0.803 \\
FE & i,j I(n\_ij) & i,j I(n\_ij) & xDeveloped i,j I(n\_ij) & xDeveloped \\
controls & & X & & X \\
\hline\hline
\multicolumn{5}{l}{\small Standard errors in parentheses} \\
\end{tabular}

```

9.4 Other Issues

The authors have significantly improved the README. As noted in the report, I encountered difficulties when running several of the Matlab scripts. This might partly be due to the limited computational capacity of the Nuvolos applications (at least for the Monte Carlo script), it would be helpful if the authors could verify the other two scripts, as they were able to generate some output but crashed afterwards.

10 Classification

As stated earlier in the report I was not able to complete the necessary scripts for the data preparation and was therefore not able to produce any output.

- ☐ full reproduction
- ☐ full reproduction with minor issues
- ☒ partial reproduction (see above)
- ☐ not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

- ☐ None.
- ☒ **Discrepancy in output** (either figures or numbers in tables or text differ)
- ☒ **Bugs in code** that were fixable by the replicator (but should be fixed in the final deposit)
- ☐ **Code missing**, in particular if it prevented the replicator from completing the reproducibility check
 - ☐ **Data preparation code missing** should be checked if the code missing seems to be data preparation code
- ☐ **Code not functional** is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- ☐ **Software not available to replicator** may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- ☐ **Insufficient time available to replicator** is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- ☐ **Data missing** is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- ☐ **Data not available** is marked when data requires additional access steps, for instance purchase or application procedure.
- ☐ **Missing README** is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements